



MPU6050 works on +3V to +5V power supply. In this circuit it is connected to +5V pin of Arduino Uno. Its serial clock (SCL) and serial data (SDA) pins are used for I2C communication between the module and Arduino Uno for providing clock pulse and for transferring data.

The mercury tilt switch module, shown in Fig. 6, can sense when it is tilted at an angle. It works because the mercury ball flows into the gap between two electrodes and closes the circuit when the module is tilted at an angle. The switch has pins for Gnd, Vcc, and Output. Its operating voltage is 5V.

The tilt switch should be placed in the iron box in such a way that it switches off when iron is placed in vertical position. The three pins of the module should be extended to the circuit.

Single-pole, double-throw (SPDT) high-level trigger AC relay board, shown in Fig. 7, can be controlled directly via a microcontroller and switch up to 10A at 250V AC. The inputs of the relay module are isolated from 230V mains to protect the delicate control circuitry. The relay output pins are connected to 3-pin 250V, 6A electrical socket.



respectively. The interrupt (INT) pin of Arduino is an output pin used to indicate that new data is available to be read from MPU6050.

SOCKET

MPU6050 module is placed inside or outside the iron box at a suitable position. Five wires are used to connect the module to Arduino Uno. Whenever the iron box moves during ironing, the positional data changes according to the movement of the iron box in different directions. The changes are analysed in the Arduino Uno board and, according to the program, the output relay energises and switches on the electric iron.

In this project, the iron's position is tracked in x, y, and z direc-

Fig. 5: Output sample values of MPU6050 module

